

Mathematical Tools

Forces (and Fields) are **vector** quantities..

We need to know how to handle them

🍁 Vector algebra

🍁 Vector calculus

We need to develop some mathematical formulations to work with...

... specially some general tools of vector calculus!

🍁 Gauss' theorem

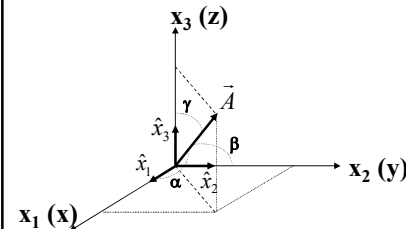
🍁 Stoke's theorem

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Mathematical Tools

I assume that you know..

🍁 what a vector is?



$$\begin{aligned}\vec{A} &= A_{x_1} \hat{x}_1 + A_{x_2} \hat{x}_2 + A_{x_3} \hat{x}_3 \\ \vec{A} &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \\ \vec{A} &: (A_x, A_y, A_z) \\ &\quad (A \cos \alpha, A \cos \beta, A \cos \gamma)\end{aligned}$$

$$\begin{aligned}\text{Magnitude of } \vec{A} \\ |\vec{A}| &= \sqrt{A_x^2 + A_y^2 + A_z^2}\end{aligned}$$

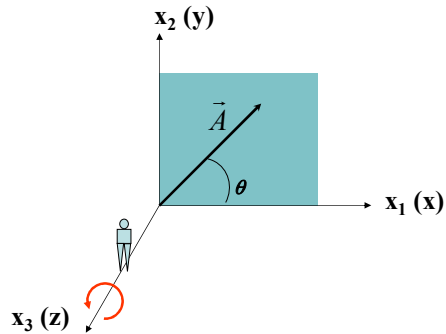
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Mathematical Tools

I assume that you know..

🍁 what a vector is?

$$\begin{aligned}\vec{A} &: (A_x, A_y) \\ &\quad (A \cos \theta, A \sin \theta)\end{aligned}$$



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Mathematical Tools

I assume that you know..

🍁 what a vector is?

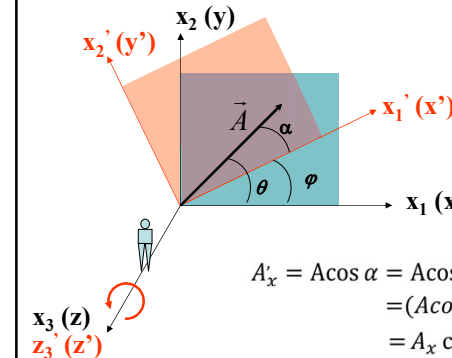
$$\begin{aligned}\vec{A} &: (A_x, A_y) \\ &\quad (A \cos \theta, A \sin \theta)\end{aligned}$$

$$x_1 \rightarrow x_1'$$

$$x_2 \rightarrow x_2'$$

$$\begin{aligned}\vec{A} &: (A'_x, A'_y) \\ &\quad (A \cos \alpha, A \sin \alpha)\end{aligned}$$

$$\begin{aligned}A'_x &= A \cos \alpha = A \cos(\theta - \varphi) \\ &= (A \cos \theta) \cos \varphi + (A \sin \theta) \sin \varphi \\ &= A_x \cos \varphi + A_y \sin \varphi\end{aligned}$$

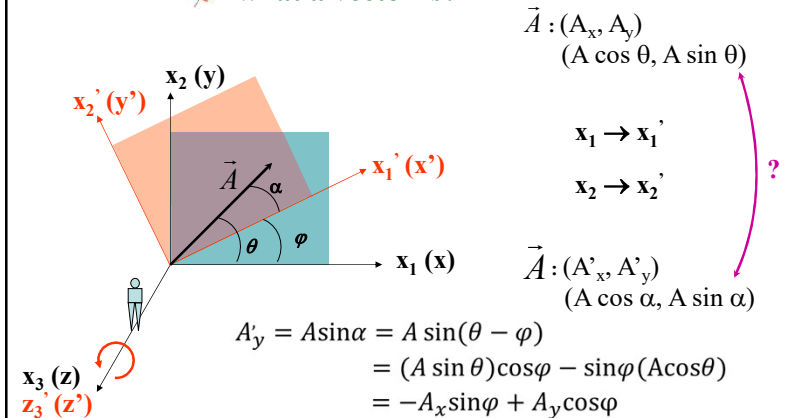


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Mathematical Tools

I assume that you know..

what a vector is?

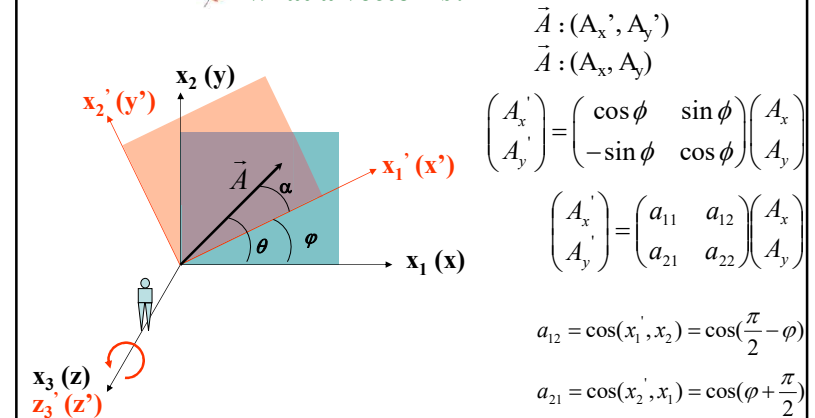


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Mathematical Tools

I assume that you know..

what a vector is?



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Mathematical Tools

I assume that you know..

what a vector is?

Must have a unique magnitude

Must have a unique direction

Components must follow the transformation relation

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Mathematical Tools

I assume that you know..

what a vector is?

vector algebra

dot (scalar) product

cross (vector) product

scalar triple product

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

vector triple product

$$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

line, surface & volume integration

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Mathematical Tools

I shall start with vector calculus...

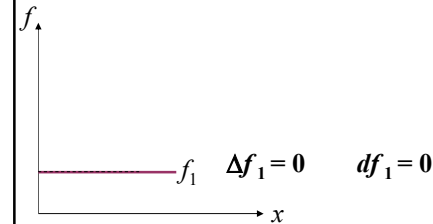
...first, vector differentiation!

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Function of One Variable: Ordinary Derivative

Consider a function f , which is a linear function of a single variable x .

Question: What would be change in f for a given change in the variable x ? $\Delta f = f(x + \Delta x) - f(x)$

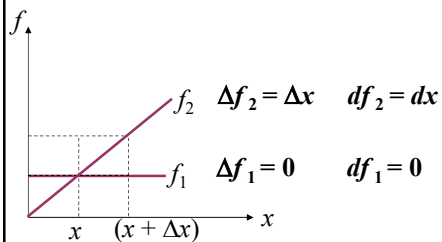


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Function of One Variable: Ordinary Derivative

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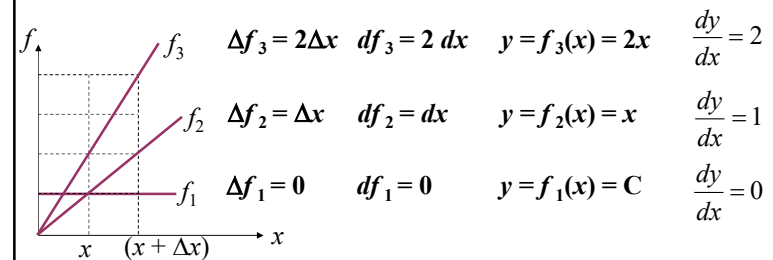


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Function of One Variable: Ordinary Derivative

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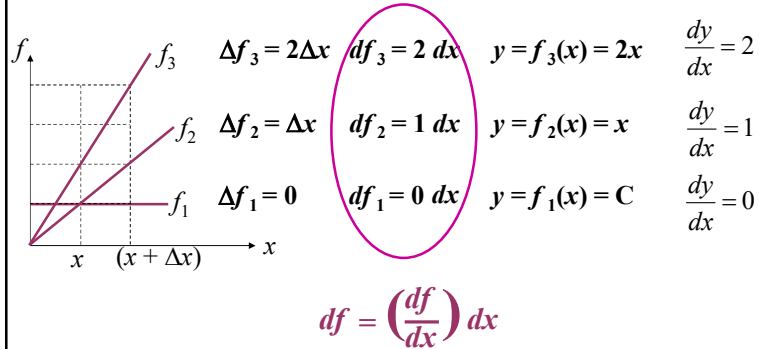


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Function of One Variable: Ordinary Derivative

Consider a function f , which is a linear function of a single variable x .

Question: What would be change in f for a given change in the variable x ? $\Delta f = f(x + \Delta x) - f(x)$

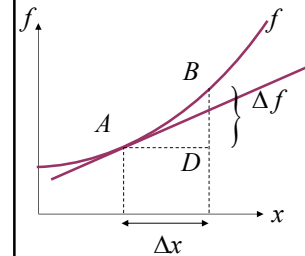


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Function of One Variable: Ordinary Derivative

Consider a function f , which is a non-linear function of a single variable x .

Question: What would be change in f for a given change in the variable x ? $\Delta f = f(x + \Delta x) - f(x)$



$$df = \left(\frac{df}{dx}\right) dx$$



Is this definition universal?

Yes, if $\Delta x \rightarrow 0$

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Function of Many Variable: Partial Derivative

Consider a function f , which is now a function of three variables x, y, z .

Example: $T(x, y, z)$: Essentially a scalar point function

Question: How fast T varies as we move from one point to another?

Note: derivative gave information about how fast the function changes if we move a little distance.....

..... but the situation here is complicated

change straight depends on which direction and how rapidly
going the horizontal point (i.e., not changing direction we are
moving) any other direction $\rightarrow$ rate of change would be intermediate

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Function of Many Variable: Partial Derivative

Consider a function f , which is now a function of three variables x, y, z .

Example: $T(x, y, z)$: Essentially a scalar point function

Question: How fast T varies as we move from one point to another?

Answer: it depends on which direction we are moving

..... It has infinite many answers, one for each direction

It looks like, finding change in $T(x, y, z)$ is too complicated?

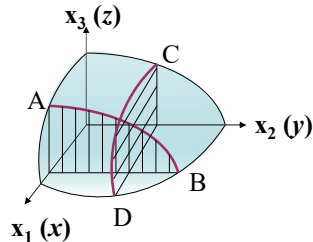
It is not

We need to know function of multi-variables more deeply and how to handle them?

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Function of Many Variable: Partial Derivative

Consider a simple case, $z = f(x, y)$ essentially a two dimensional case.



$$\frac{\partial z}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{z(x, y + \Delta y) - z(x, y)}{\Delta y}$$

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{z(x + \Delta x, y) - z(x, y)}{\Delta x}$$

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Function of Many Variable: Partial Derivative

Example of $z = f(x, y)$: $f(x, y) = 2x^3y^2 + y^3$

$$\frac{\partial f}{\partial x} = ?$$

$$\frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x} = ?$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y} = ?$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \frac{\partial f}{\partial x} = ?$$

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Function of Many Variable: Partial Derivative

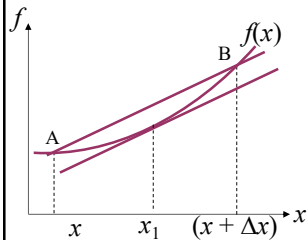
Come back to the original question: change in $f(x, y, z)$?

$$\Delta f(x, y) = f(x + \Delta x, y + \Delta y) - f(x, y)$$

$$= f(x + \Delta x, y + \Delta y) - f(x + \Delta x, y) + f(x + \Delta x, y) - f(x, y)$$

Using the Mean Value Theorem of calculus

$$f(x + \Delta x) - f(x) = \Delta x f'(x_1)$$



$$\Delta f(x, y) = \frac{\partial f(x_1, y)}{\partial x} \Delta x + \frac{\partial f(x + \Delta x, y_1)}{\partial y} \Delta y$$

for $\Delta x, \Delta y \rightarrow 0$

$$\Delta f(x, y) = \frac{\partial f(x, y)}{\partial x} \Delta x + \frac{\partial f(x, y)}{\partial y} \Delta y$$

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Function of Many Variable: Partial Derivative

$$\Delta f(x, y) = \frac{\partial f(x, y)}{\partial x} \Delta x + \frac{\partial f(x, y)}{\partial y} \Delta y$$

Extending to three dimension

$$\begin{aligned} df(x, y, z) &= \frac{\partial f(x, y, z)}{\partial x} \Delta x + \frac{\partial f(x, y, z)}{\partial y} \Delta y + \frac{\partial f(x, y, z)}{\partial z} \Delta z \\ df(x, y, z) &= \left(\frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z} \right) \cdot (\Delta x \hat{x} + \Delta y \hat{y} + \Delta z \hat{z}) \\ &= (\vec{\nabla} f) \cdot (d\vec{r}) \end{aligned}$$

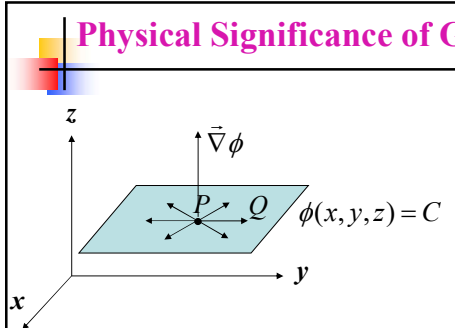
$\vec{\nabla} f$ is defined as Gradient of a scalar function f .

$$\vec{\nabla} \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \quad \vec{\nabla} f \neq f \vec{\nabla}$$

$\vec{\nabla}$ (scalar field) $\longrightarrow$ (vector field)

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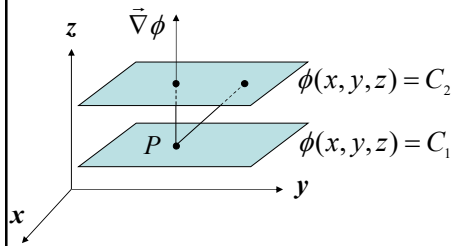
Physical Significance of Gradient



$$d\phi = \vec{\nabla} \phi \cdot d\vec{r} = 0$$

$\vec{\nabla} \phi \perp d\vec{r}$

Universal Fact

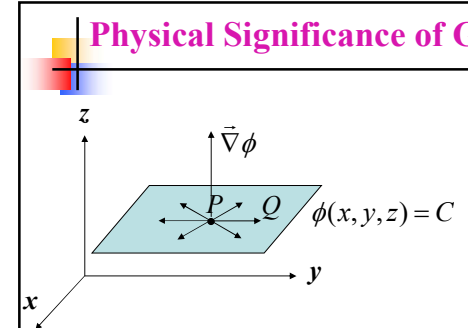


$$d\phi = C_2 - C_1 = \vec{\nabla} \phi \cdot d\vec{r}$$

For a given $d\phi$, $|d\vec{r}|$ is min in the direction of $\vec{\nabla} \phi$

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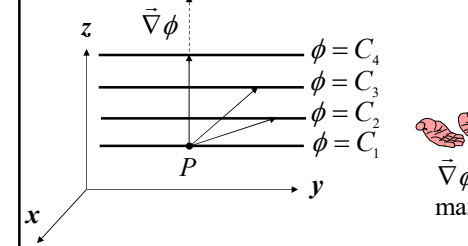
Physical Significance of Gradient



$$d\phi = \vec{\nabla} \phi \cdot d\vec{r} = 0$$

$\vec{\nabla} \phi \perp d\vec{r}$

Universal Fact



Let's fix, $|d\vec{r}|$

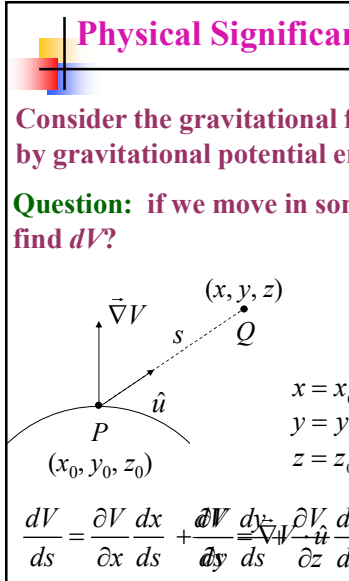
$\vec{\nabla} \phi$ indicates the direction of maximum change in ϕ

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Physical Significance of Gradient

Consider the gravitational field as the scalar field represented by gravitational potential energy (V). $V = f(x, y, z) = f(s)$

Question: if we move in some other direction, how should we find dV ?



$$\hat{u} = a\hat{x} + b\hat{y} + c\hat{z}$$

$$PQ = s\hat{u} = as\hat{x} + bs\hat{y} + cs\hat{z} = (x-x_0)\hat{x} + (y-y_0)\hat{y} + (z-z_0)\hat{z}$$

$$\left. \begin{aligned} x &= x_0 + as \\ y &= y_0 + bs \\ z &= z_0 + cs \end{aligned} \right\} \text{Parametric equation of the line PQ}$$

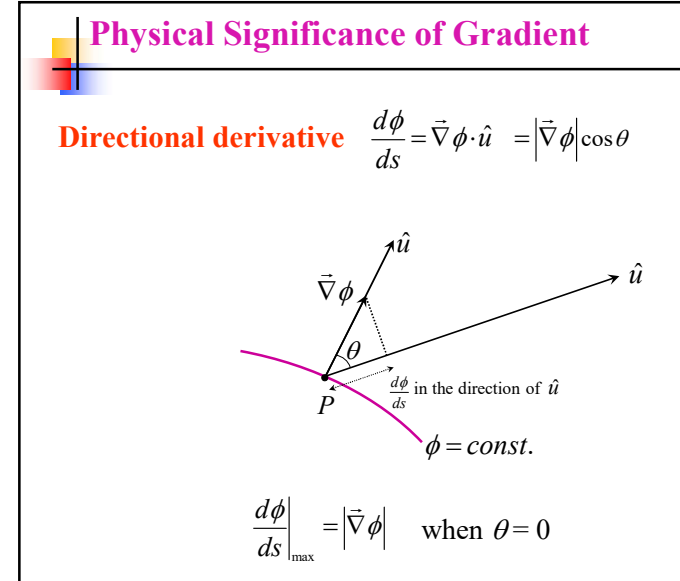
$$\frac{dV}{ds} = \frac{\partial V}{\partial x} \frac{dx}{ds} + \frac{\partial V}{\partial y} \frac{dy}{ds} + \frac{\partial V}{\partial z} \frac{dz}{ds} = \vec{\nabla} V \cdot \hat{u}$$

Directional derivative

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Physical Significance of Gradient

Directional derivative $\frac{d\phi}{ds} = \vec{\nabla} \phi \cdot \hat{u} = |\vec{\nabla} \phi| \cos \theta$



$\frac{d\phi}{ds} \Big|_{\max} = |\vec{\nabla} \phi|$ when $\theta = 0$

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Physical Significance of Gradient

Example: Find the directional derivative of $f(x, y) = x^2y + xz$ at $(1, 2, -1)$ in the direction of $\vec{A} = 2\hat{x} - 2\hat{y} + \hat{z}$.

$$\vec{\nabla}f = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) (x^2y + xz)$$

$$= \hat{x}(2xy + z) + x^2\hat{y} + x\hat{z}$$

$$\hat{u} = \frac{2\hat{x} - 2\hat{y} + \hat{z}}{\sqrt{4+4+1}} = \frac{1}{3}(2\hat{x} - 2\hat{y} + \hat{z})$$

$$\vec{\nabla}f \cdot \hat{u} \Big|_{(1,2,-1)} = \frac{2}{3}(2xy + z) - \frac{2}{3}x^2 + \frac{1}{3}x \Big|_{(1,2,-1)} = \frac{5}{3} \quad \blacktriangleleft$$

$$\vec{\nabla}f \cdot \hat{u} \Big|_{(1,2,-1)}$$

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Physical Significance of Gradient

Example: Find $\vec{\nabla}f$ where $f(x, y, z) = r = \sqrt{x^2 + y^2 + z^2}$.

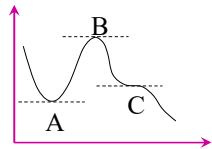
Answer: $\vec{\nabla}r = \hat{r} \quad \blacktriangleleft$

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Physical Significance of Gradient

What if Gradient vanishes? $\vec{\nabla}\phi = 0$

Recall: $df/dx = 0 \rightarrow$ stationary points (A, B, C)



A $\rightarrow$ local minima ($f'' > 0$)

B $\rightarrow$ local maxima ($f'' < 0$)

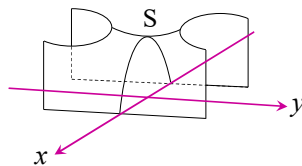
C $\rightarrow$ point of inflection ($f'' = 0$)

$\vec{\nabla}\phi = 0 \rightarrow$ stationary points

local minima

local maxima

saddle point (S)



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The del operator

$$\vec{\nabla} \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

is an operator which “acts upon” a given function.

Possible ways it can act upon:

on a scalar function ϕ : $\vec{\nabla}\phi$ (Gradient)

on a vector function $\vec{V}$: $\vec{\nabla} \cdot \vec{V}$ (Divergence)

$\vec{\nabla} \times \vec{V}$ (Curl)

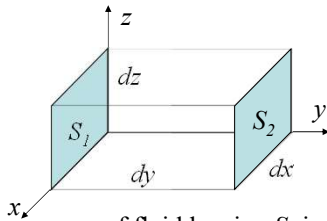
Divergence

$$\begin{aligned} \vec{\nabla} \cdot \vec{V} &= \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot (\hat{x}V_x + \hat{y}V_y + \hat{z}V_z) \\ &= \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \end{aligned}$$

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Divergence: Physical Significance

Consider a region in which a compressible fluid $\rho(x,y,z)$ is flowing with a velocity $\vec{v} = v_x(x,y,z)\hat{x} + v_y(x,y,z)\hat{y} + v_z(x,y,z)\hat{z}$



mass of fluid entering through S_1 in a time interval $dt = \text{density} \times \text{volume}$

$$= (\rho) \Big|_{y=0} dx dz (v_y) \Big|_{y=0} dt$$

$$= (\rho v_y) \Big|_{y=0} dx dz dt$$

mass of fluid leaving S_2 in a time interval $dt = (\rho) \Big|_{y=dy} dx dz (v_y) \Big|_{y=dy} dt$

$$= (\rho v_y) \Big|_{y=dy} dx dz dt$$

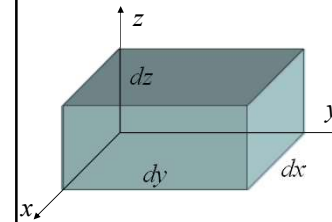
Net mass of fluid leaving the volume element in a time interval dt along y-direction: $= [(\rho v_y) \Big|_{y=dy} - (\rho v_y) \Big|_{y=0}] dx dz dt$

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Divergence: Physical Significance

Net mass of fluid leaving the volume element in a time interval dt along y-direction: $= [(\rho v_y) \Big|_{y=dy} - (\rho v_y) \Big|_{y=0}] dx dz dt = \frac{\partial(\rho v_y)}{\partial y} dy dx dz dt$

Net mass of fluid leaving the volume element in a time interval dt along x-direction: $= \frac{\partial(\rho v_x)}{\partial x} dx dz dy dt$



Net mass of fluid leaving the volume element in a time interval dt along z-direction: $= \frac{\partial(\rho v_z)}{\partial z} dz dx dy dt$

Net mass of fluid leaving the volume element in a time interval dt :

$$= \left(\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z} \right) dy dx dz dt$$

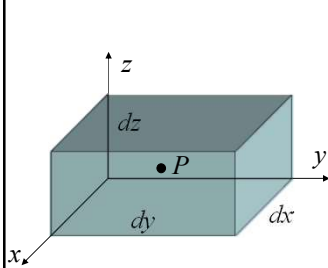
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Divergence: Physical Significance

Net flow out per unit time:

$$= \left(\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z} \right) dy dx dz = \vec{\nabla} \cdot (\rho \vec{v}) dx dy dz$$

Net flow out per unit time per unit volume: $= \vec{\nabla} \cdot (\rho \vec{v})$



Divergence is the measure of how much a vector field spreads out (diverges) from a given point

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Divergence: Continued

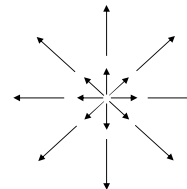
$$\vec{\nabla} \cdot (\vec{v}) = (\text{net mass left} - \text{net mass entered}) / \text{volume} / \text{time}$$

Divergence measures flux of the vector field

$$\vec{\nabla} \cdot (\vec{v}) = 0 \rightarrow \text{Vector field is solenoidal}$$

Examples

Large + ve divergence



$$\vec{V} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

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Divergence: Continued

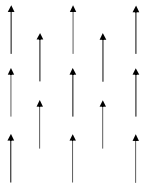
$$\vec{\nabla} \cdot (\vec{v}) = (\text{net mass left} - \text{net mass entered})/\text{volume/time}$$

Divergence measures flux of the vector field

$$\vec{\nabla} \cdot (\vec{v}) = 0 \rightarrow \text{Vector field is solenoidal}$$

Examples

zero divergence



$$\vec{V} = c\hat{z}$$

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(c) = 0$$

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Divergence: Continued

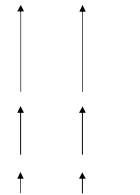
$$\vec{\nabla} \cdot (\vec{v}) = (\text{net mass left} - \text{net mass entered})/\text{volume/time}$$

Divergence measures flux of the vector field

$$\vec{\nabla} \cdot (\vec{v}) = 0 \rightarrow \text{Vector field is solenoidal}$$

Examples

finite divergence



$$\vec{V} = z\hat{z}$$

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 1$$

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Divergence: Continued

$$\vec{\nabla} \cdot (\vec{v}) = (\text{net mass left} - \text{net mass entered})/\text{volume/time}$$

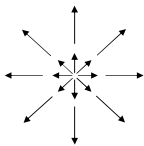
Divergence measures flux of the vector field

$$\vec{\nabla} \cdot (\vec{v}) = 0 \rightarrow \text{Vector field is solenoidal}$$

Examples

$$\vec{V} = \frac{\hat{r}}{r^2}$$

large divergence



$$\vec{\nabla} \cdot \vec{V} = \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) = \vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right) = \vec{\nabla} \cdot \left(\frac{x\hat{x} + y\hat{y} + z\hat{z}}{r^3} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right)$$

$$\frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) = \frac{\partial}{\partial x} \left[x(x^2 + y^2 + z^2)^{-3/2} \right] = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} - \frac{3}{2} \frac{x(2x)}{(x^2 + y^2 + z^2)^{5/2}}$$

$$= \frac{1}{r^3} - \frac{3x^2}{r^5} + \frac{1}{r^3} - \frac{3y^2}{r^5} + \frac{1}{r^3} - \frac{3z^2}{r^5} = \frac{3}{r^3} - \frac{3(x^2 + y^2 + z^2)}{r^5} = 0?$$

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Curl

$$\vec{\nabla} \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\vec{V} = \hat{x}V_x + \hat{y}V_y + \hat{z}V_z$$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_x & V_y & V_z \end{vmatrix}$$

$$= \hat{x} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) + \hat{y} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) + \hat{z} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right)$$

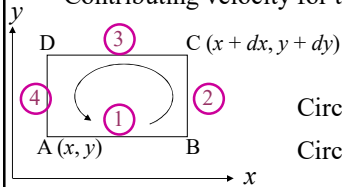
$$\vec{\nabla} \times \vec{V} \neq \vec{V} \times \vec{\nabla}$$

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Curl: Physical Significance

Consider the circulation of a fluid around a differential loop in the xy -plane.

Contributing velocity for this circulation: $\vec{v} = v_x(x, y)\hat{x} + v_y(x, y)\hat{y}$



Circulation along path "1" $= v_x(x, y)dx$

Circulation along path "2" $= v_y(x+dx, y)dy$

Circulation along path "3" $= -v_x(x, y+dy)dx$

Circulation along path "4" $= -v_y(x, y)dy$

Circulation along ABCDA

$$= v_x(x, y)dx + v_y(x+dx, y)dy - v_x(x, y+dy)dx - v_y(x, y)dy$$

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Curl: Physical Significance

Circulation along ABCDA

$$= v_x(x, y)dx + v_y(x+dx, y)dy - v_x(x, y+dy)dx - v_y(x, y)dy$$

Using Taylor expansion $v_y(x+dx, y) = v_y(x, y) + \frac{\partial v_y}{\partial x}dx$

Circulation along ABCDA

$$= v_x(x, y)dx + v_y(x, y)dy + \frac{\partial v_y}{\partial x}dxdy - v_x(x, y)dx - \frac{\partial v_x}{\partial y}dxdy - v_y(x, y)dy$$

$$= \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) dxdy = (\vec{\nabla} \times \vec{v})_z dxdy$$



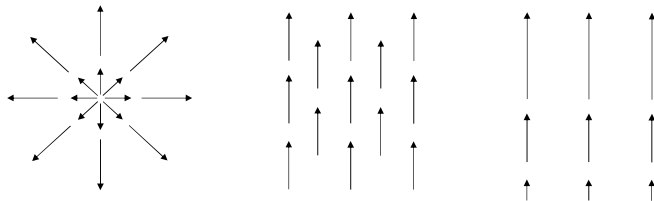
Curl of a vector field is the measure of the circulation of the field in a particular plane

How much a field curls around the point in question

$$\vec{\nabla} \times \vec{v} = 0 \rightarrow \text{Irrotational Field}$$

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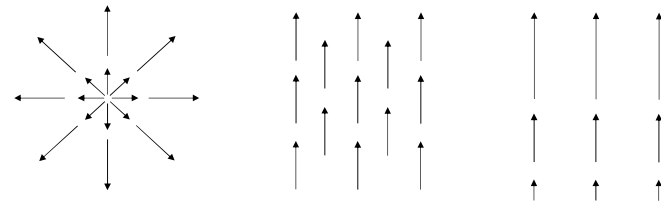
Curl: Example



$$\vec{\nabla} \times \vec{v} = 0 \rightarrow \text{Irrotational Fields}$$

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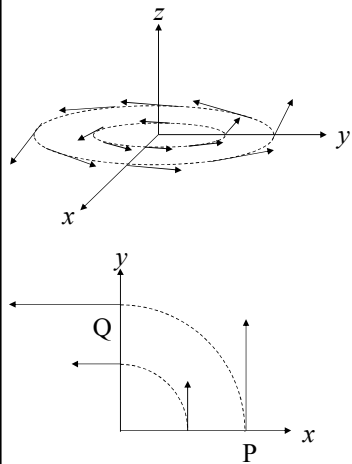
Curl: Example



Home Work: Calculate curl for each field from first principle

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Curl: Example



finite divergence

At P ($x = x, y = 0$): $x\hat{y}$

At Q ($x = 0, y = y$): $-y\hat{x}$

$$\vec{V} = -y\hat{x} + x\hat{y}$$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix}$$

$$= 2\hat{z}$$